SWARM – RSCL Outdoor Formation experiment

Objective

Fly the four X500 V2 drones outdoors, streaming RTK-grade pose into ROS 2 and obeying velocity commands, so we can run and test a leader–follower formation controller in the field.

Formation Controller

A leader–follower controller where each follower estimates the leader's velocity and acceleration from pose alone (a sliding-mode differentiator), then holds a 3D, possibly rotating, offset from it. It already works in simulation and indoors under motion capture (MOCAP).

Deliverables

The outdoor replacement for the MOCAP room: every drone's pose delivered to his code on ROS 2 using RTK and F9P GPS, and his velocity commands executed on the drones. We do not write the formation controller.

Hardware

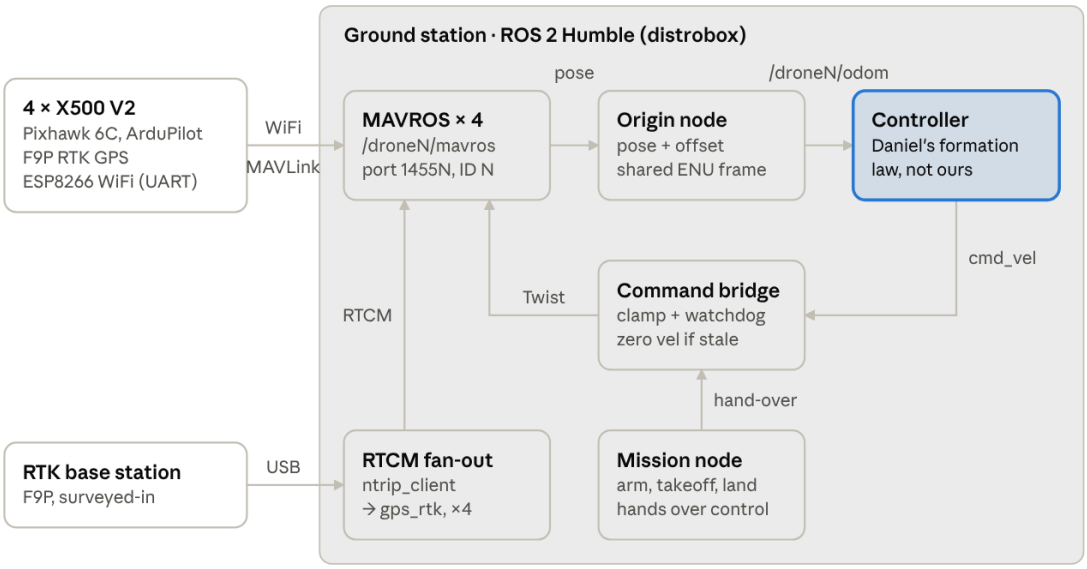
4 × Holybro X500 V2 quadcopters, Pixhawk 6C, ArduPilot (ArduCopter), each with an F9P RTK receiver and an ESP8266 WiFi bridge joined to the router `vanttec_drones_swarm`. RTK base: Holybro H-RTK F9P Base. Ground station: Ubuntu 22.04 host running ROS 2 Humble.

Deliverables

#	Requirement	Done when	Difficulty
1	ROS 2 Humble on the ground station	`ros2 multicast receive` works between distrobox and a second laptop; MAVROS installed from apt	Low
2	MAVROS over WiFi, one-to-many	4 MAVROS instances under `/drone1` … `/drone4`, each `connected: true` on its `state` topic, for 10 minutes with all 4 powered.	Medium
3	Full pose per drone on `nav_msgs/Odometry`	`/droneN/odom` publishes position + quaternion at the agreed rate (target ≥ 20 Hz)	Low
4	One shared local origin	4 drones placed on known marks on the ground: their published positions match the tape-measured offsets within 5 cm. Process data using ROS services to give an adequate position	High
5	Tested velocity interface (`geometry_msgs/Twist`)	Each drone, in GUIDED, follows a scripted Twist sequence in SITL and outdoors, it holds position within 3 s when commands stop or it may land.	Medium.
6	Version control + documentation	One GitHub repo, a README	Low

Architecture

The code to be used should only ever see two topics per drone: an `Odometry` coming in and a `Twist` going out. Everything else (MAVLink, WiFi, RTK, origin math) stays inside our stack in the ground station.



The diagram shows **how data moves between the drones and the formation controller.** Everything inside the grey box runs on our ground station laptop. There are three flows.

1. Pose loop (top row, left to right): "where is each drone?"

- **4 × X500 V2:** each drone's Pixhawk computes its own position (from the F9P GPS) and orientation, and sends them over WiFi through the ESP8266 using MAVLink, ArduPilot's message language.
- **MAVROS × 4:** one translator per drone. It turns MAVLink into ROS 2 topics. Each drone uses its own port (14551–14554) and ID (1–4) so the four don't mix.
- **Origin node**:: each drone measures position from its own private zero point. This node shifts all four into **one shared frame**, so their positions can be compared.
- It publishes `/droneN/odom` → **Controller**, its colored blue as it is not within the project scope.

2. Command loop (right side, going back): "where should each drone go?"

- **Controller** → **`cmd_vel`:** flight controller outputs a velocity command per drone.
- **Command bridge**: checks each command, limits the speed (clamp), and if commands stop arriving it sends zero velocity so the drone hovers (watchdog).
- → **Twist** → **MAVROS** → **WiFi** → **drone:** the drone flies at that velocity in GUIDED mode.

3. RTK corrections (bottom left): “make the GPS accurate to centimeters”

- **RTK base station** (your H-RTK F9P Base) sits still on a tripod and measures GPS error.
- **USB → RTCM fan-out:** we read its correction data (RTCM) and copy it to all four drones through MAVROS. That’s what turns 1–3 m GPS into ~1 cm that is basically using RTK fixed, instead of just the GPS.

The mission node

It handles the flight sequence: arm, take off, then **hand over** control to the command bridge, and land at the end.

The key idea: if someone wants to test itl only ever sees `odom` going in and `cmd_vel`. If we build it right, his code can’t tell it’s outdoors instead of the MOCAP room.

Topic	Type	Direction	Frame (coordinate reference system)	Rate
`/droneN/odom`	`nav_msgs/Odometry`	us → formation controller	`frame_id: swarm_origin` (ENU, shared), `child_frame_id: droneN/base_link` (FLU)	≥ 20 Hz target
`/droneN/cmd_vel`	`geometry_msgs/Twist`	Formation controller → us	linear in swarm ENU (m/s), `angular.z` = yaw rate (rad/s)	10–50 Hz
`/droneN/status`	small custom msg or `mavros_msgs/State`	us → formation controller		1–5 Hz

Inside our stack, the per-drone pipeline is: MAVROS publishes `/droneN/mavros/local_position/odom` (ENU, relative to that drone's own EKF origin) and `/droneN/mavros/global_position/gp_origin`. Our **origin node** shifts the pose into `swarm_origin` and republishes `/droneN/odom`.

Our **command bridge** validates each `Twist`, clamps it, and forwards it to `/droneN/mavros/setpoint_velocity/cmd_vel_unstamped`.

Note: a velocity is the same in every frame that shares axes, so the origin shift only touches pose, never commands. Both must stay in ENU world axes

ENU = East, North, Up. It's a way to describe a position as "how many meters from a chosen zero point" along three fixed directions:

- **x = East** (+ is east, − is west)
- **y = North** (+ is north, − is south)
- **z = Up** (+ is up, − is down)

Why it matters for us: two conventions meet in our system.

- **ROS uses ENU.** Our `/droneN/odom` and `cmd_vel` are ENU.
- **ArduPilot uses NED = North, East, Down.** x and y are swapped, and z points down, so 10 m altitude is z = −10.
- **FLU = Forward, Left, Up**, is the same idea but attached to the drone and turning with it. It's used for the drone's own axes (`child_frame_id` in the odom message)
- ArduPilot's body version is FRD (Forward, Right, Down), and MAVROS converts that too.

MAVROS converts between them automatically

Technical findings and decisions

ArduPilot will not let us force a common EKF origin while using GPS, so the shared frame must be computed on the ground station.

1. MAVROS, not ArduPilot's native DDS

MAVROS 2, installed with `sudo apt install ros-humble-mavros ros-humble-mavros-extras` plus the GeographicLib datasets script (mavros repo).

2. Network: ¿How to connect each drone ?

1. **Router side:** connect the ASUS router, you will se the network: vanttec_droneswarm. Connect with the password: droneswarm!. Afterwards enter to the admin page, to do so execute this command on terminal to check the routers ip.`ip route | grep default` . Afterwards enter to a browser paste the ip number and the put the wifi network and password as username and password. As you enter navigate to LAN on the left side panel, and go to DHCP server, you will see manual ip assignation is enabled with mac address, and probably drones will already be added.

   THe reason I used static ip, is because using mac address is easier than assigning an ip form the esp side. Using the mac is much safer if the esp gets flashed or changed parameters by accident.

   To check the mac adress of your drone you can execute this command in the therminal:

   `sudo arp-scan --localnet`

   That will scan the network and show all the ips, to check which is the drone, connect it and disconnect it so you can check which one it is. After you have the mac, you can go back to the router admin page and add it putting the mac and an ip, I recommend using 192.168.50.n this much be smaller than 192.168.50.100 as this is the starting range for dynamic ip assignment so consider that before assigning, you must apply changes. Also be shure to save the ip in a text or note so you can identify your drone in further steps.

   Thats it you now have added your drone a static ip. congrats

   NOte: when you light up a drone and it connects to the wifi its light turns yellow and starts blinking, if your drone doesnt do that it means probably something is wrong, it may be connected so you can check using scan localnet but it may not be sending information due to baud misunderstanding or similar, check all the parameters or even reflash the esp if necessary.

2. **Per drone (Pixhawk side):** verify parameters unique `SYSID_THISMAV`, must be the same as the label in the drone. On the TELEM port the ESP8266 is wired to TELEM2 , `SERIALx_PROTOCOL = 2` (MAVLink 2) and `SERIALx_BAUD = 500000.` Baud must be the same between esp and telem port

- **Per drone (ESP8266 side):** to enter an ESP parameter it exists an integrated API server with this format. You just need to paste this in the browser replacing it with the ip you assigned before, `http://<Esp ip>/getparameters`

You will enter a page in which you can add the parameters and set up them yourself, nevertheless I have found this page may not apply changes correctly but it is friendly to check the parameters here. Paste this link in the browser replacing with the IP of the drone ESP and everything will be setted up. Also change the hport to a unique port 1455N, change the N for a value above 14554

http://<IP_ESP>/setparameters?ipsta=0.0.0.0&gatewaysta=0.0.0.0&subnetsta=0.0.0.0&hport=1455N&cport=14555&ssid=vt_droneXX_ap&pwd=droneswarm%21&reboot=1

ArduPilot's mavesp8266 firmware in station mode (`WIFI_MODE = 1`) on `vanttec_drones_swarm`, a static IP (`WIFI_IPSTA`, `WIFI_GATEWAYSTA`, `WIFI_SUBNETSTA`, all three together), and `WIFI_UDP_HPORT = 1455N` instead of the default 14550 (mavesp8266 parameters). UART stays at its baud rate 500000 default.

- The ESP8266 is a 2.4 GHz, telemetry-class link. Keep streams minimal, put the router on a fixed, quiet 2.4 GHz channel, and give the ground station a fixed IP. The ESP also answers as its own MAVLink component (ID 240); that is normal.
- Optional: each MAVROS node's `gcs_url` can forward to QGroundControl for live safety monitoring.

3. Message rates: request exactly what we need

PROPOSED BY CLAUDE !!!!, revise carefully

Use `SET_MESSAGE_INTERVAL` per message rather than `SRx_` groups; it gives precise control and saves bandwidth (ArduPilot data requests). Ask for `LOCAL_POSITION_NED` (32), `ATTITUDE_QUATERNION` (31) and `GLOBAL_POSITION_INT` (33) at 20–25 Hz each. If rates keep reverting, a GCS is overriding them: disable QGC's stream-rate control or set the TELEM port's "ignore streamrate" option (same page).

4. Shared origin: computed on the ground station

PROPOSED BY CLAUDE !!!!, revise carefully

- ArduPilot's local frame is relative to the EKF origin, set where the vehicle first got a good position (Guided commands). Once set, it cannot be moved (Home and EKF origin), and with GPS it is set automatically. Forum threads confirm unifying origins across a swarm is not possible from the vehicle side (discussion).
- **Decision (method A):** choose one `swarm_origin` (the RTK base station's surveyed position). Read each drone's `gp_origin` once, convert it to an ENU offset from `swarm_origin`

(GeographicLib or `pymap3d.geodetic2enu`), and add that constant offset to the drone's local pose.
- **Cross-check (method B):** convert each drone's `GLOBAL_POSITION_INT` straight to ENU about `swarm_origin`. Publish both during testing; they should agree within centimeters.
- **Altitude:** with RTK fixed, test using GPS as the EKF height source (`EK3_SRC1_POSZ`) so barometer drift does not differ between drones.
- **Yaw:** RTK fixes position, not heading. Heading comes from each compass, so compass error becomes formation yaw error. Calibrate all four compasses on the field day.

5. RTK corrections into all four drones

PROPOSED BY CLAUDE !!!!, revise carefully

- ArduPilot accepts corrections from a local base through the ground station over MAVLink, from an NTRIP service, or by radio to the GPS (RTK correction page). The base must survey-in first (2 m accuracy is common, same page).
- **Decision:** base F9P on the ground station by USB → `ntrip_client`'s serial-device launch publishes `mavros_msgs/RTCM` (ntrip_client) → fanned out to each `/droneN/mavros/gps_rtk/send_rtcm` (mavros_extras `gps_rtk` plugin, which sends `GPS_RTCM_DATA`) (mavros_extras). Fallback: QGC or Mission Planner injection.
- Hardware: each drone needs an F9P-class receiver. Holybro's H-RTK F9P is rated 0.01 m in RTK with up to 8 Hz RTK output (Holybro); the EKF fills the gaps between fixes with the IMU.

6. Velocity commands in GUIDED mode, with our own watchdog

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- MAVROS `setpoint_velocity` takes a `Twist` in ENU and converts it to NED; its default frame is `LOCAL_NED` (plugin source). ArduPilot then executes `SET_POSITION_TARGET_LOCAL_NED` velocity + yaw rate.
- ArduPilot stops a vehicle after 3 s without a new velocity command and expects at least 1 Hz (Guided commands). Our bridge streams at ≥ 10 Hz and sends zero velocity if Daniel's command is older than 0.5 s; after 2 s it switches the drone to LOITER.
- Flight sequence owned by our mission node: GUIDED → arm → takeoff → hand control to Daniel → land. Speed clamp for first flights, geofence on, battery failsafe on, one RC safety pilot per drone.

7. Simulation before every field step

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`sim_vehicle.py -v Copter --count 4 --auto-sysid --auto-offset-line 90,10` starts 4 copters with unique IDs (Using SITL). The offset spawn gives each copter a different EKF origin, which is exactly the case method A must fix, so origin math can be tested before any drone flies.

Learning resources

Topic	Resource	Why it matters here
Quaternions	3Blue1Brown + Ben Eater interactive lessons (intro)	Read Daniel's slides 4–8 and our Odometry messages without guessing
RTK	SparkFun: What is GPS RTK?	Why base + rover reaches about 1 cm
ROS 2 basics refresher	ROS 2 Humble tutorials	Namespaces, launch files, remapping, parameters
ArduPilot + ROS overview	ArduPilot ROS page	
MAVROS 2	mavros repo (ros2 branch) and launch files on ROS index	
WiFi bridge	mavesp8266 parameters	Port and baud configuration of each drone's link
Message rates	Requesting data from the autopilot	20–25 Hz pose without flooding WiFi
EKF origin rules	Home and EKF origin	Why we compute the shared frame on the ground
Geodetic → ENU	pymap3d (`geodetic2enu`)	The core math of the origin node
Quaternion math, deep	Solà, Quaternion kinematics (arXiv)	Reference when frames or signs look wrong
ArduPilot RTK setup	RTK GPS correction (fixed baseline)	Survey-in, injection methods, RTK float vs fixed
ROS 2 RTCM source	ntrip_client	Serial base station → `mavros_msgs/RTCM`
Guided commands	Copter commands in Guided mode	Frames, type masks, the 3 s stop rule; MAVProxy test commands
MAVROS velocity plugin	setpoint_velocity source	How the Twist becomes MAVLink
Multi-vehicle SITL	Using SITL	`--count 4 --auto-sysid` swarm in one command

Sources

- ArduPilot: Copter commands in Guided mode
- ArduPilot: Setting Home and/or EKF origin
- ArduPilot: Requesting data from the autopilot
- ArduPilot: RTK GPS correction (fixed baseline)
- ArduPilot: ESP32 WiFi telemetry
- ArduPilot: Using SITL
- ArduPilot wiki source: ROS 1 / ROS 2
- ArduPilot forum: unifying EKF origins across vehicles
- MAVROS repository · mavros on ROS index · mavros_extras
- MAVROS setpoint_velocity plugin source
- ntrip_client on ROS index
- Holybro DroneCAN H-RTK F9P Rover
- Ubuntu: ROS 2 networking and multicast
- pymap3d
- SparkFun: What is GPS RTK?
- Solà, Quaternion kinematics for the error-state Kalman filter
- Quaternions explorable (eater.net)
- ArduPilot: ESP8266 WiFi telemetry
- mavesp8266 parameters (ArduPilot fork)
- Holybro H-RTK F9P Base

Lista de tópicos

/diagnostics

/drone1/mavros/mavros/adsb/send

/drone1/mavros/mavros/adsb/vehicle

/drone1/mavros/mavros/battery

/drone1/mavros/mavros/cam_imu_sync/cam_imu_stamp

/drone1/mavros/mavros/camera/image_captured

/drone1/mavros/mavros/cellular_status/status

/drone1/mavros/mavros/companion_process/status

/drone1/mavros/mavros/esc_status/info

/drone1/mavros/mavros/esc_status/status

/drone1/mavros/mavros/esc_telemetry/telemetry

/drone1/mavros/mavros/estimator_status

/drone1/mavros/mavros/extended_state

/drone1/mavros/mavros/fake_gps/mocap/pose

/drone1/mavros/mavros/geofence/fences

/drone1/mavros/mavros/gimbal_control/device/attitude_status

/drone1/mavros/mavros/gimbal_control/device/info

/drone1/mavros/mavros/gimbal_control/device/set_attitude

/drone1/mavros/mavros/gimbal_control/manager/info

/drone1/mavros/mavros/gimbal_control/manager/set_attitude

/drone1/mavros/mavros/gimbal_control/manager/set_manual_control

/drone1/mavros/mavros/gimbal_control/manager/set_pitchyaw

/drone1/mavros/mavros/gimbal_control/manager/status

/drone1/mavros/mavros/global_position/compass_hdg

/drone1/mavros/mavros/global_position/global

/drone1/mavros/mavros/global_position/gp_lp_offset

/drone1/mavros/mavros/global_position/gp_origin

/drone1/mavros/mavros/global_position/local

/drone1/mavros/mavros/global_position/raw/fix

/drone1/mavros/mavros/global_position/raw/gps_vel

/drone1/mavros/mavros/global_position/raw/satellites

/drone1/mavros/mavros/global_position/rel_alt

/drone1/mavros/mavros/global_position/set_gp_origin

/drone1/mavros/mavros/gps_input/gps_input

/drone1/mavros/mavros/gps_rtk/rtk_baseline

/drone1/mavros/mavros/gps_rtk/send_rtcm

/drone1/mavros/mavros/gpsstatus/gps1/raw

/drone1/mavros/mavros/gpsstatus/gps1/rtk

/drone1/mavros/mavros/gpsstatus/gps2/raw

/drone1/mavros/mavros/gpsstatus/gps2/rtk

/drone1/mavros/mavros/home_position/home

/drone1/mavros/mavros/home_position/set

/drone1/mavros/mavros/imu/data

/drone1/mavros/mavros/imu/data_raw

/drone1/mavros/mavros/imu/diff_pressure

/drone1/mavros/mavros/imu/mag

/drone1/mavros/mavros/imu/static_pressure

/drone1/mavros/mavros/imu/temperature_baro

/drone1/mavros/mavros/imu/temperature_imu

/drone1/mavros/mavros/landing_target/lt_marker

/drone1/mavros/mavros/landing_target/pose

/drone1/mavros/mavros/landing_target/pose_in

/drone1/mavros/mavros/local_position/accel

/drone1/mavros/mavros/local_position/odom

/drone1/mavros/mavros/local_position/pose

/drone1/mavros/mavros/local_position/pose_cov

/drone1/mavros/mavros/local_position/velocity_body

/drone1/mavros/mavros/local_position/velocity_body_cov

/drone1/mavros/mavros/local_position/velocity_local

/drone1/mavros/mavros/log_transfer/raw/log_data

/drone1/mavros/mavros/log_transfer/raw/log_entry

/drone1/mavros/mavros/mag_calibration/report

/drone1/mavros/mavros/mag_calibration/status

/drone1/mavros/mavros/manual_control/control

/drone1/mavros/mavros/manual_control/send

/drone1/mavros/mavros/mission/reached

/drone1/mavros/mavros/mission/waypoints

/drone1/mavros/mavros/mocap/pose

/drone1/mavros/mavros/mocap/tf

/drone1/mavros/mavros/mount_control/command

/drone1/mavros/mavros/mount_control/orientation

/drone1/mavros/mavros/mount_control/status

/drone1/mavros/mavros/nav_controller_output/output

/drone1/mavros/mavros/obstacle/send

/drone1/mavros/mavros/obstacle_distance_3d/send

/drone1/mavros/mavros/odometry/in

/drone1/mavros/mavros/odometry/out

/drone1/mavros/mavros/onboard_computer/status

/drone1/mavros/mavros/open_drone_id/basic_id

/drone1/mavros/mavros/open_drone_id/operator_id

/drone1/mavros/mavros/open_drone_id/self_id

/drone1/mavros/mavros/open_drone_id/system

/drone1/mavros/mavros/open_drone_id/system_update

/drone1/mavros/mavros/optical_flow/ground_distance

/drone1/mavros/mavros/optical_flow/raw/optical_flow

/drone1/mavros/mavros/optical_flow/raw/send

/drone1/mavros/mavros/param/event

/drone1/mavros/mavros/play_tune

/drone1/mavros/mavros/radio_status

/drone1/mavros/mavros/rallypoint/rallypoints

/drone1/mavros/mavros/rangefinder/rangefinder

/drone1/mavros/mavros/rangefinder_pub

/drone1/mavros/mavros/rangefinder_sub

/drone1/mavros/mavros/rc/in

/drone1/mavros/mavros/rc/out

/drone1/mavros/mavros/rc/override

/drone1/mavros/mavros/setpoint_accel/accel

/drone1/mavros/mavros/setpoint_attitude/cmd_vel

/drone1/mavros/mavros/setpoint_attitude/thrust

/drone1/mavros/mavros/setpoint_position/global

/drone1/mavros/mavros/setpoint_position/global_to_local

/drone1/mavros/mavros/setpoint_position/local

/drone1/mavros/mavros/setpoint_raw/attitude

/drone1/mavros/mavros/setpoint_raw/global

/drone1/mavros/mavros/setpoint_raw/local

/drone1/mavros/mavros/setpoint_raw/target_attitude

/drone1/mavros/mavros/setpoint_raw/target_global

/drone1/mavros/mavros/setpoint_raw/target_local

/drone1/mavros/mavros/setpoint_trajectory/desired

/drone1/mavros/mavros/setpoint_trajectory/local

/drone1/mavros/mavros/setpoint_velocity/cmd_vel

/drone1/mavros/mavros/setpoint_velocity/cmd_vel_unstamped

/drone1/mavros/mavros/sim_state/acceleration

/drone1/mavros/mavros/sim_state/attitude

/drone1/mavros/mavros/sim_state/global_position

/drone1/mavros/mavros/sim_state/velocity_body

/drone1/mavros/mavros/sim_state/velocity_local

/drone1/mavros/mavros/state

/drone1/mavros/mavros/status_event

/drone1/mavros/mavros/statustext/recv

/drone1/mavros/mavros/statustext/send

/drone1/mavros/mavros/sys_status

/drone1/mavros/mavros/terrain/data

/drone1/mavros/mavros/terrain/report

/drone1/mavros/mavros/terrain/request

/drone1/mavros/mavros/time_reference

/drone1/mavros/mavros/timesync_status

/drone1/mavros/mavros/trajectory/desired

/drone1/mavros/mavros/trajectory/generated

/drone1/mavros/mavros/trajectory/path

/drone1/mavros/mavros/tunnel/in

/drone1/mavros/mavros/tunnel/out

/drone1/mavros/mavros/vfr_hud

/drone1/mavros/mavros/vision_pose/pose

/drone1/mavros/mavros/vision_pose/pose_cov

/drone1/mavros/mavros/vision_speed/speed_twist

/drone1/mavros/mavros/vision_speed/speed_twist_cov

/drone1/mavros/mavros/vision_speed/speed_vector

/drone1/mavros/mavros/wind_estimation

/move_base_simple/goal

/parameter_events

/rosout

/tf

/tf_static

/uas1/mavlink_sink

/uas1/mavlink_source

Lista de servicios

/drone1/mavros/mavros/adsb/describe_parameters

/drone1/mavros/mavros/adsb/get_parameter_types

/drone1/mavros/mavros/adsb/get_parameters

/drone1/mavros/mavros/adsb/list_parameters

/drone1/mavros/mavros/adsb/set_parameters

/drone1/mavros/mavros/adsb/set_parameters_atomically

/drone1/mavros/mavros/cam_imu_sync/describe_parameters

/drone1/mavros/mavros/cam_imu_sync/get_parameter_types

/drone1/mavros/mavros/cam_imu_sync/get_parameters

/drone1/mavros/mavros/cam_imu_sync/list_parameters

/drone1/mavros/mavros/cam_imu_sync/set_parameters

/drone1/mavros/mavros/cam_imu_sync/set_parameters_atomically

/drone1/mavros/mavros/camera/describe_parameters

/drone1/mavros/mavros/camera/get_parameter_types

/drone1/mavros/mavros/camera/get_parameters

/drone1/mavros/mavros/camera/list_parameters

/drone1/mavros/mavros/camera/set_parameters

/drone1/mavros/mavros/camera/set_parameters_atomically

/drone1/mavros/mavros/cellular_status/describe_parameters

/drone1/mavros/mavros/cellular_status/get_parameter_types

/drone1/mavros/mavros/cellular_status/get_parameters

/drone1/mavros/mavros/cellular_status/list_parameters

/drone1/mavros/mavros/cellular_status/set_parameters

/drone1/mavros/mavros/cellular_status/set_parameters_atomically

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/drone1/mavros/mavros/cmd/command

/drone1/mavros/mavros/cmd/command_int

/drone1/mavros/mavros/cmd/describe_parameters

/drone1/mavros/mavros/cmd/get_parameter_types

/drone1/mavros/mavros/cmd/get_parameters

/drone1/mavros/mavros/cmd/land

/drone1/mavros/mavros/cmd/land_local

/drone1/mavros/mavros/cmd/list_parameters

/drone1/mavros/mavros/cmd/set_home

/drone1/mavros/mavros/cmd/set_parameters

/drone1/mavros/mavros/cmd/set_parameters_atomically

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/drone1/mavros/mavros/cmd/takeoff_local

/drone1/mavros/mavros/cmd/trigger_control

/drone1/mavros/mavros/cmd/trigger_interval

/drone1/mavros/mavros/cmd/vtol_transition

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/drone1/mavros/mavros/companion_process/get_parameter_types

/drone1/mavros/mavros/companion_process/get_parameters

/drone1/mavros/mavros/companion_process/list_parameters

/drone1/mavros/mavros/companion_process/set_parameters

/drone1/mavros/mavros/companion_process/set_parameters_atomically

/drone1/mavros/mavros/describe_parameters

/drone1/mavros/mavros/distance_sensor/describe_parameters

/drone1/mavros/mavros/distance_sensor/get_parameter_types

/drone1/mavros/mavros/distance_sensor/get_parameters

/drone1/mavros/mavros/distance_sensor/list_parameters

/drone1/mavros/mavros/distance_sensor/set_parameters

/drone1/mavros/mavros/distance_sensor/set_parameters_atomically

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/drone1/mavros/mavros/esc_status/set_parameters_atomically

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/drone1/mavros/mavros/fake_gps/set_parameters_atomically

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/drone1/mavros/mavros/geofence/push

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/drone1/mavros/mavros/gimbal_control/manager/get_info

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/drone1/mavros/mavros/log_transfer/raw/log_request_list

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/drone1/mavros/mavros/mag_calibration/list_parameters

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/drone1/mavros/mavros/vision_speed/get_parameter_types

/drone1/mavros/mavros/vision_speed/get_parameters

/drone1/mavros/mavros/vision_speed/list_parameters

/drone1/mavros/mavros/vision_speed/set_parameters

/drone1/mavros/mavros/vision_speed/set_parameters_atomically

/drone1/mavros/mavros/wind/describe_parameters

/drone1/mavros/mavros/wind/get_parameter_types

/drone1/mavros/mavros/wind/get_parameters

/drone1/mavros/mavros/wind/list_parameters

/drone1/mavros/mavros/wind/set_parameters

/drone1/mavros/mavros/wind/set_parameters_atomically

/drone1/mavros/mavros_node/describe_parameters

/drone1/mavros/mavros_node/get_parameter_types

/drone1/mavros/mavros_node/get_parameters

/drone1/mavros/mavros_node/list_parameters

/drone1/mavros/mavros_node/set_parameters

/drone1/mavros/mavros_node/set_parameters_atomically

/drone1/mavros/mavros_router/add_endpoint

/drone1/mavros/mavros_router/del_endpoint

/drone1/mavros/mavros_router/describe_parameters

/drone1/mavros/mavros_router/get_parameter_types

/drone1/mavros/mavros_router/get_parameters

/drone1/mavros/mavros_router/list_parameters

/drone1/mavros/mavros_router/set_parameters

/drone1/mavros/mavros_router/set_parameters_atomically